

第九章

方向梯度和导数

梯度: $\nabla u = (\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z})$ 最大方向

方向导数 $\vec{n} = (x_0, y_0, z_0)$

$$\frac{\partial u}{\partial n} \Big|_{(x_0, y_0, z_0)} = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right) \Big|_{(x_0, y_0, z_0)} \cdot \left(\frac{x_0}{r}, \frac{y_0}{r}, \frac{z_0}{r} \right)$$

多元函数微分学的几何应用

参数方程切线 $\begin{cases} x(t) \\ y(t) \\ z(t) \end{cases} \Rightarrow (x'(t), y'(t), z'(t))$

$$x'(x-x_0) + y'(y-y_0) + z'(z-z_0) = 0 \quad \frac{x-x_0}{x'(t_0)} + \frac{y-y_0}{y'(t_0)} + \frac{z-z_0}{z'(t_0)} = 0$$

曲线切平面与法线 $f(x, y, z) = 0$

$$\vec{n} = \left(\frac{\partial f}{\partial x} \Big|_{(x_0, y_0, z_0)}, F_1(x_0, y_0, z_0), F_2(x_0, y_0, z_0) \right)$$

一阶方程的总切曲线切线

$$\vec{n} = (F_x(x_0, y_0, z_0), F_y(x_0, y_0, z_0), F_z(x_0, y_0, z_0))$$

$$\vec{n} = (G_x(x_0, y_0, z_0), G_y(x_0, y_0, z_0), G_z(x_0, y_0, z_0))$$

$$\vec{n} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} i & j & k \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}$$

切平面

线性化与全微分定义 $f(x, y) = f_x(x_0, y_0)(x-x_0) + f_y(x_0, y_0)(y-y_0)$

$$\Delta z = f_x(x_0, y_0) \Delta x + f_y(x_0, y_0) \Delta y \quad \text{全微分}$$

极值与最值

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \Rightarrow (x_0, y_0) \text{ 可疑点 (驻点)}$$

$$A = f_{xx}(x_0, y_0) \quad B = f_{xy}(x_0, y_0) \quad C = f_{yy}(x_0, y_0)$$

$$AC - B^2 > 0 \quad \begin{cases} A > 0 \text{ 极小} \\ A < 0 \text{ 极大} \end{cases}$$

$AC - B^2 = 0$ 无法判定

$AC - B^2 < 0$ 非极值

最值代入边界并 $g(x, y) = 0$ 确定并定范围在这个范围下找之最
边界与中心比较得最值

条件极值.

方法1: 代入法. 将该条件代入并确定范围后再求极值

方法2: 拉格朗日乘数法.

$$\begin{cases} f_x(x, y) + \lambda g_x(x, y) = 0 \\ f_y(x, y) + \lambda g_y(x, y) = 0 \\ g(x, y) = 0 \end{cases}$$

求驻点 找出可疑点后可直接代入过 A, B, C 验证,

第十章

重积分 { 估值不等式: $m\sigma \leq \iint f(x,y) dx dy \leq M\sigma$
积分中值定理: $\iint f(x,y) d\sigma = f(\xi, \eta)\sigma$

二重积分

意义: 二维物品质量面积 三维体积

如何积分: X型区域 先积Y J型区域 先积X

$$\iint f(x,y) d\sigma = \int_a^b dx \int_c^d f(x,y) dy$$

通过 首次积分 选择好积的

极坐标下: $\iint f(x,y) d\sigma = \int d\theta \int f(r\cos\theta, r\sin\theta) r dr$

应用: 曲面积分: $ds = \sqrt{1+z_x'^2+z_y'^2} dx$
 $= \sqrt{x'^2+y'^2} dt$

$$\text{质心: } \bar{x} = \frac{\iint x \rho d\sigma}{\iint \rho d\sigma}$$

$$\text{转动惯量: } I_y = \iint x^2 \rho(x,y) dx dy$$

三重积分

意义: 三维物品质量, 体积

如何积分: 先Z后 - $\int dz \iint f(x,y,z) d\sigma$

$$Z \text{ 后} = \iint d\sigma \int f(x,y,z) dz$$

极坐标: $\iiint f(x,y,z) dv = \int_0^{2\pi} d\theta \int_0^{\alpha} d\phi \int_{r(\phi,\theta)}^{R(\phi,\theta)} f(r\cos\phi, r\sin\phi, r) r^2 dr$

柱坐标系: $\int dz \int d\phi \int f(r\cos\phi, r\sin\phi, z) r dr$

对称性: 关于XOY对称 若 $f(x,y,z) = -f(x,y,-z) \Rightarrow \iiint f(x,y,z) = 0$
 $f(x,y,z) = f(x,y,-z) \Rightarrow 2 \iiint_V f(x,y,z)$

关于YZ对称 若 $f(x,-y,z) = -f(x,y,z) \Rightarrow \iiint f(x,y,z) = 0$
若 X轴 1, Y轴 2, Z轴 3 自后往前 若有轴对称1.

$$\text{应用: 质心: } \bar{x} = \frac{\iiint x \rho dv}{\iiint \rho dv}$$

$$\text{转动惯量: } I_x = \iiint (y^2+z^2) \rho dv$$

第十一章

场的基本概念

数量场, 向量场, 散度场, 梯度场

数量场求梯度 $\longrightarrow$ 散度场;

向量场求散度 $\longrightarrow$ 数量场.

向量场求旋度 $\longrightarrow$ 向量场.

数量场曲线积分. (第一类曲线)

$$ds = \sqrt{1+y'^2} dx = \sqrt{1+x'^2} dy = \sqrt{x'^2 + y'^2} dt = \sqrt{\dot{x}^2 + \dot{y}^2} dt$$

$$ds = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt$$

数量场曲面积分

$$\iint_S f(x, y, z) ds = \iint_{D_{xy}} f(x, y, z(x, y)) \sqrt{1+z_x^2+z_y^2} dx dy$$

向量场曲线积分. (第二类曲线)

$$W = \int \vec{F} \cdot \vec{\tau} ds = \int P dx + Q dy + R dz$$

化成单变量后积分.

$$\text{环流: } \int_C \vec{F} \cdot \vec{\tau} ds = \int_C P dx + Q dy$$

$$\text{通量: } \int_C \vec{F} \cdot \vec{n} ds = \int_C P dy - Q dx$$

格林定理

$$\text{环流 } \oint \vec{F} \cdot \vec{\tau} ds = \int_C P dx + Q dy = \oint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx$$

$$\text{通量 } \oint \vec{F} \cdot \vec{n} ds = \int_C P dy - Q dx = \oint \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx$$

高斯定理和斯托克斯公式

$$\text{通量 } \oint \vec{F} \cdot \vec{n} ds = \iiint_V \nabla \cdot \vec{F} dV$$

$$\oint \vec{F} \cdot \vec{\tau} ds = \iint_S \nabla \times \vec{F} \cdot d\vec{S}$$

$$\int a^x dx = a^x \ln a$$

$$\int e^x dx = e^x$$

$$\int \sec^2 x dx = \tan x$$

$$\int \csc^2 x dx = -\cot x$$

$$\int \frac{1}{1+x^2} dx = \arctan x$$

$$\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan \frac{x}{a}$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin \frac{x}{a}$$

$$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln |x + \sqrt{x^2-a^2}|$$

$$\int \frac{1}{\sqrt{x^2+a^2}} dx = \ln |x + \sqrt{x^2+a^2}|$$

